

DEEPAKRAJ B

PYTHON BACKEND DEVELOPER | FASTAPI, DJANGO & CLOUD

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PROFILE

Python Backend Developer with hands-on experience building scalable RESTful APIs, microservices, and automated data processing platforms. Skilled in **Python (3.8+)**, **FastAPI**, **Django**, and **Frappe Framework**, with strong expertise in database design and query optimization (**PostgreSQL**, **MySQL**), caching strategies (**Redis**), and containerized deployment on **AWS (EC2, S3, Docker)**. Proven track record delivering production backend systems, AI query integration tools (**LangGraph**, **LangChain**, **NL2SQL**), and enterprise software solutions.

PROFESSIONAL EXPERIENCE

Python Software Engineer | Kritilabs Technologies Pvt. Ltd., Chennai, India **July 2024 - Present**

Natural Language SQL Query Engine & AWS Deployment **Python, FastAPI, LangGraph, Docker, AWS**

- Built and deployed a production **NL2SQL Engine** using **Python**, **LangGraph**, and **FastAPI**, enabling natural language database querying with multi-step validation and SQL query execution.
- Designed backend workflow logic handling tool execution, error handling, and query verification to ensure consistent query output.
- Implemented vector-based context retrieval and state memory using **LangGraph** and **ChromaDB**, reducing prompt overhead and improving response speeds.
- Containerized Python services using **Docker** and established automated **CI/CD pipelines** for deployment on **AWS (EC2, S3)**.

Manufacturing ERP Platform: Python / Frappe Engineering **Python, Frappe, MySQL, REST APIs**

- Engineered backend modules in **Python / Frappe Framework** for an ERPNext manufacturing platform serving a 700-employee enterprise across 4 production facilities (BOM management, work orders, and inventory tracking).
- Developed **FastAPI webhooks** and REST API integrations connecting legacy ERP, inventory, and mobile barcode scanners with Frappe APIs, reducing manual data entry by **80%**.
- Optimized **MySQL & PostgreSQL** database schemas, indexes, and queries, improving REST API response times by **30%** under heavy load.

Data Processing Automation & AI Analytics Backend **Python, PostgreSQL, Async Services**

- Developed an automated Python data processing tool querying multiple relational databases to generate automated executive reporting metrics for leadership teams.
- Built backend services for a crowd monitoring analytics system deployed for **Indian Railways** and real-time vehicle fleet tracking APIs for **IOCL & BPCL**.

TECHNICAL SKILLS

Programming Languages: Python (3.8+), SQL, JavaScript, HTML5/CSS3

Python Web Frameworks: FastAPI, Django, Django REST (DRF), Frappe, Flask

AI & Query Tools: LangGraph, LangChain, NL2SQL, ChromaDB

Cloud & DevOps: AWS (EC2, S3), Docker, CI/CD, Nginx, Linux

Databases & Caching: PostgreSQL, MySQL, Redis, SQLAlchemy, Django ORM

Database Engineering: Query Optimization, Indexing, Schema Design

API Design & Security: RESTful APIs, Microservices, Async, JWT, OAuth2

Testing & Tooling: Pytest, Unittest, Postman, OpenAPI, Git, JIRA

EDUCATION

B.Tech in Computer Science & Engineering | SASTRA Deemed University, Thanjavur, Tamil Nadu **CGPA: 7.2955** | 2020 - 2024

CERTIFICATIONS

- Python Crash Course - Google
- Machine Learning Specialization
- Software Engineering Specialization - IBM
- Backend Development Specialization - Meta